QSD Supplement S2: Energy Scale and Dressing

From Planck Mass to Electroweak Scale

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Abstract

This supplement derives the QSD energy scale hierarchy in full detail. Starting from the Planck mass M_P as the sole dimensionful input, we construct the electroweak scale through three independently motivated steps: information reduction by the factor 2^{-60} (from the code state space), geometric enhancement by φ^2 (from the C_5 Laplacian eigenvalue ratio), and strain correction by $(1 - 6\varsigma)$ (from the K_4/C_5 mismatch). The resulting dimensionful mass $\mu_0 = 25.34$ GeV is compared to the independently derived dimensionless VEV $\nu_0 = 25.18$, yielding the dressing factor $\xi = 1.00636$ GeV. We present tree-level and corrected values for all predicted masses, and derive the Higgs mass, syndrome energy, and W boson mass from the radion scale.

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1 The Hierarchy Problem

The Standard Model contains two fundamental scales:

$$M_{\mathrm{P}} = \sqrt{\frac{\hbar c^5}{G}} = 1.2209 \times 10^{19}\ \mathrm{GeV} \qquad \text{(gravity)} \tag{1}$$

$$v_{\mathrm{EW}} = (\sqrt{2}\, G_F)^{-1/2} = 246.22\ \mathrm{GeV} \qquad \text{(electroweak)} \tag{2}$$

Their ratio is enormous:

$$\frac{M_{\mathrm{P}}}{v_{\mathrm{EW}}} = 4.96 \times 10^{16} \tag{3}$$

In the Standard Model, maintaining this hierarchy requires cancellations between radiative corrections to 17 decimal places—the *hierarchy problem* [1, 2]. QSD resolves this: the ratio is not fine-tuned but is fixed by the information content of the $[[5, 1, 3]]$ quantum error-correcting code.

1.1 Strategy

The derivation proceeds in three steps, each traceable to the axioms of S1:

1. **Information reduction** ($\nabla \cdot 2^{60}$): The code's state space volume sets the number of microscopic configurations per macroscopic state.
2. **Geometric enhancement** ($\times \varphi^2$): The C_5 eigenvalue ratio amplifies the base scale to the radion scale.
3. **Strain correction** ($\times (1 - 6\varsigma)$): The K_4/C_5 mismatch reduces the effective coupling on each edge.

No additional parameters are introduced at any stage.

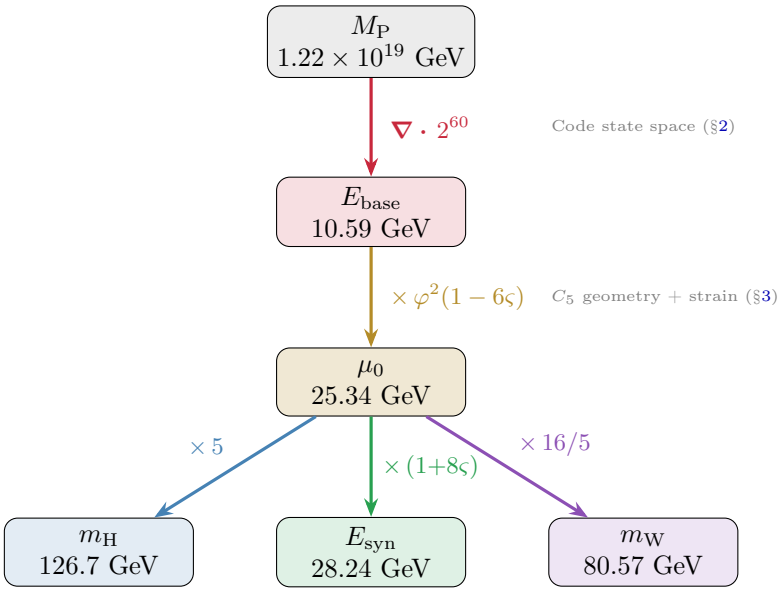


Figure 1: The complete hierarchy of QSD energy scales. Every arrow is derived from the code structure; no free parameters enter.

2 The Information Reduction Factor 2^{-60}

2.1 The State Space Volume

Theorem 2.1 (State Space Volume)**.** *The total number of distinguishable microscopic configurations of the K_4 code structure is:*

$$\boxed{\mathcal{N} = 16^{15} = 2^{60} = 1.152\,921\,504\,606\,847 \times 10^{18}} \tag{4}$$

Proof. We derive the two factors independently.

Factor 1: Hilbert space dimension $D = 16$. The bulk consists of 4 qubits (Axiom 1 of S1) [3]. The Hilbert space is $\mathcal{H}_4 = (\mathbb{C}^2)^{\otimes 4}$, with dimension:

$$D = \dim(\mathcal{H}_4) = 2^4 = 16 \tag{5}$$

Factor 2: Independent interaction channels $N = 15$. The space of all operators on $\mathcal{H}_4$ has dimension $D^2 = 256$. Removing the identity (which generates no dynamics), the number of independent interaction channels is:

$$N = D^2 - 1 = 256 - 1 = 15 \tag{6}$$

This equals $\dim(\mathfrak{su}(4))$, the dimension of the Lie algebra of the natural symmetry group $SU(4)$ acting on 4 qubits. Equivalently, $N = 15$ can be decomposed as:

$$15 = 5 \times 3 \tag{7}$$

where 5 is the number of physical qubits in the code and 3 is the number of Pauli operators per qubit (X, Y, Z).

State space volume. Each of the $N = 15$ channels can independently be in any of $D = 16$ states. The total volume is:

$$\mathcal{N} = D^N = 16^{15} = (2^4)^{15} = 2^{60} \tag{8}$$

Numerical value:

$$2^{60} = 1\,152\,921\,504\,606\,846\,976 \approx 1.153 \times 10^{18} \tag{9}$$

□

Remark 2.2 (Interpretation). The factor 2^{60} counts the number of microscopic configurations of the code that map to a single macroscopic (observable) state. This is the information-theoretic origin of the hierarchy: the electroweak scale is "diluted" relative to the Planck scale by the enormous number of microstates per macrostate.

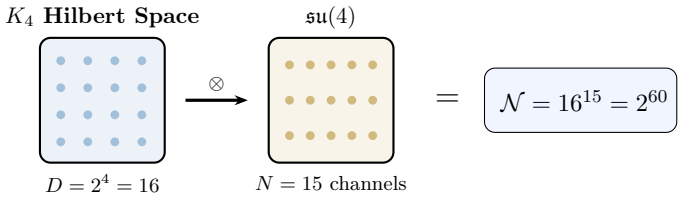


Figure 2: The state space volume arises from 16 quantum states distributed across 15 independent interaction channels.

2.2 The Base Energy Scale

Definition 2.3 (Base Scale)**.** The base energy scale is the Planck mass divided by the state space volume:

$$E_{\text{base}} = \frac{M_{\text{P}}}{\mathcal{N}} = \frac{M_{\text{P}}}{2^{60}} \tag{10}$$

Proposition 2.4 (Numerical Value)**.**

$$E_{\text{base}} = \frac{1.2209 \times 10^{19}\ GeV}{1.1529 \times 10^{18}} = 10.590\ GeV \tag{11}$$

Proof. Direct division: $1.2209/1.1529 = 1.0590$, with the exponent $10^{19}/10^{18} = 10$. □

Remark 2.5. The base scale 10.59 GeV lies in the middle of the hadron physics range—between the b-quark mass (∼4.2 GeV) and the Z boson mass (∼91 GeV). This is the "raw" Planck-reduced scale before geometric corrections are applied.

2.3 Alternative Derivation: Suppression Cascade

Proposition 2.6 (Cascade Interpretation)**.** *The factor* 2^{-60} *can be understood as 15 successive suppression events, each reducing the energy by a factor of* $1/16$*:*

$$\frac{1}{16^{15}} = \left(\frac{1}{16}\right)^{15} = \left(e^{-1/\text{Hb}_4}\right)^{15} \tag{12}$$

Proof. From S1, Corollary 2.3: $e^{-1/\text{Hb}_4} = 2^{-4} = 1/16$. Each information transfer event through the holographic screen suppresses the energy by this factor. With 15 independent channels, the total suppression is $(1/16)^{15} = 2^{-60}$. □

This cascade interpretation is physically intuitive: the Planck-scale energy must "pass through" all 15 interaction channels of the $\mathfrak{su}(4)$ algebra, losing a factor of 16 at each step, before emerging at the electroweak scale.

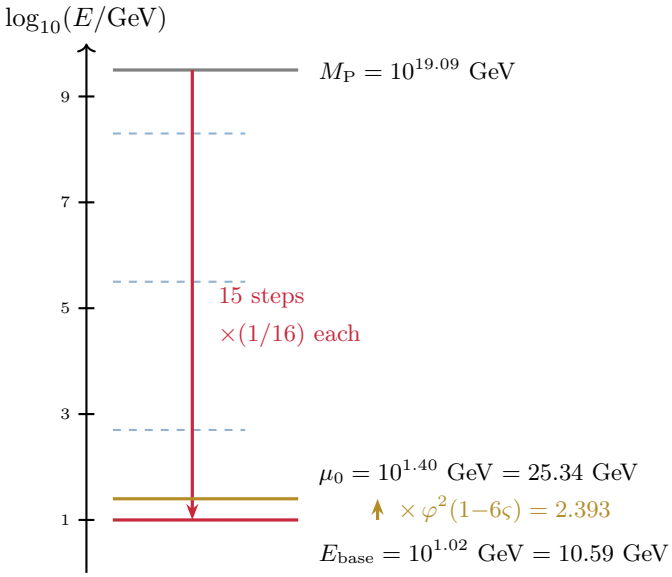


Figure 3: Log-scale view of the hierarchy. The Planck mass cascades through 15 suppression events (×1/16 each), then receives a geometric enhancement to reach the radion scale.

3 Geometric Enhancement and Strain Correction

3.1 Enhancement by φ^2

The base scale E_{base} does not yet account for the geometry of the C_5 graph. The radion field lives on the pentagon, whose eigenvalue ratio provides a multiplicative factor.

Proposition 3.1 (C_5 Enhancement)**.** *The C_5 Laplacian eigenvalue ratio (S1, Theorem 4.3) enhances the base scale:*

$$E_{\text{base}} \times \varphi^2 = 10.590 \times 2.6180 = 27.724 \ GeV \tag{13}$$

Physical interpretation: The golden ratio factor φ^2 reflects the fact that the radion (which is the angular oscillation mode of the pentagon) inherits the geometric amplification encoded in the C_5 spectrum. The ratio $\varphi^2 = \mu_2/\mu_1$ measures how much harder it is to excite the higher C_5 mode compared to the lower one, and this ratio boosts the energy of the physical excitation.

3.2 Strain Correction $(1 - 6\varsigma)$

Proposition 3.2 (Strain Reduction)**.** *Each of the 6 edges of K_4 loses a fraction ς of its effective coupling due to the K_4/C_5 incompatibility:*

$$1 - 6\varsigma = 1 - 6 \times 0.014325 = 1 - 0.08595 = 0.91405 \tag{14}$$

Physical interpretation: The 6 edges of K_4 are the interaction channels for Heisenberg exchange. But K_4 and C_5 cannot be simultaneously realized on the same set of qubits without conflict: K_4 has 6 edges while C_5 has 5, and they encode different symmetries (S_4 vs. $\mathbb{Z}_5$). Each edge pays a "tax" of ς for this incompatibility, reducing the total effective coupling by 6ς.

Remark 3.3. The factor $6\varsigma \approx 0.086$ represents an 8.6% reduction—small enough that the base-scale calculation is meaningful, but large enough to be physically consequential. This correction shifts the radion mass from 27.72 GeV down to 25.34 GeV, which propagates into all derived quantities.

4 The Radion Mass: Complete Calculation

Theorem 4.1 (Radion Mass Formula)**.** *The dimensionful radion mass is:*

$$\boxed{\mu_0 = \frac{M_{\text{P}}}{2^{60}} \times \varphi^2 \times (1 - 6\varsigma) = 25.34 \ GeV} \tag{15}$$

Proof. Assembling the three factors derived above:

Step 1: Information reduction.

$$E_{\text{base}} = \frac{M_{\text{P}}}{2^{60}} = \frac{1.2209 \times 10^{19}}{1.1529 \times 10^{18}} = 10.590 \text{ GeV} \tag{16}$$

Step 2: Geometric enhancement.

$$E_{\text{base}} \times \varphi^2 = 10.590 \times 2.6180 = 27.724 \text{ GeV} \tag{17}$$

Step 3: Strain correction.

$$27.724 \times (1 - 6\varsigma) = 27.724 \times 0.91405 = 25.342 \text{ GeV} \tag{18}$$

Final result:

$$\mu_0 = 25.34 \text{ GeV} \tag{19}$$

□

Factor	Origin	Value	Axiom (S1)
M_{P}	Gravitational constant	1.221×10^{19} GeV	Input
2^{-60}	Code state space (16^{15})	8.674×10^{-19}	Axioms 1, 2
φ^2	C_5 eigenvalue ratio	2.6180	Axiom 1
$(1-6\varsigma)$	K_4/C_5 strain	0.9141	Axioms 1, 2, 3

Table 1: Every factor in the hierarchy formula traced to its axiomatic origin.

5 The Dimensionless VEV and the Dressing Factor

5.1 Two Independent Routes to ∼25

QSD produces the number "∼25" through two entirely independent calculations:

Definition 5.1 (Dimensionless Radion VEV)**.**

$$\nu_0 = \frac{\text{Hb}_4}{\varsigma} = \frac{0.36067}{0.014325} = 25.178 \tag{20}$$

This is a *pure number*—the ratio of two dimensionless quantities (holographic bit density and geometric strain). It carries no physical units.

Definition 5.2 (Dimensionful Radion Mass)**.**

$$\mu_0 = \frac{M_{\mathrm{P}}}{2^{60}} \times \varphi^2 \times (1-6\varsigma) = 25.342 \text{ GeV} \tag{21}$$

This has units of energy and is derived from the hierarchy formula (Theorem 4.1).

5.2 The Dressing Factor ξ

Definition 5.3 (Dressing Factor)**.** The dressing factor is the ratio of the dimensionful mass to the dimensionless VEV:

$$\boxed{\xi \equiv \frac{\mu_0}{\nu_0} = \frac{25.342 \text{ GeV}}{25.178} = 1.00651 \text{ GeV}} \tag{22}$$

Proposition 5.4 (Near-Unity of ξ)**.** *The dressing factor deviates from 1 GeV by only 0.65%:*

$$\frac{|\xi - 1 \ GeV|}{1 \ GeV} = 0.0065 \tag{23}$$

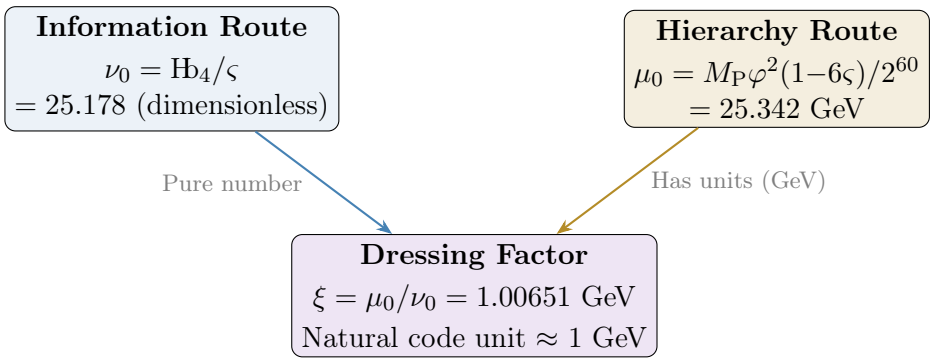


Figure 4: Two independent routes converge to the same numerical value ∼25, with the dressing factor ξ bridging the dimensionless and dimensionful descriptions.

5.3 Interpretation of ξ

The dressing factor $\xi = 1.007$ GeV has a precise physical meaning: it is the **natural energy unit of the** $[[5, 1, 3]]$ **code** when expressed in SI-derived units. The near-coincidence $\xi \approx 1$ GeV is a nontrivial statement:

> *The information-theoretic unit of the error-correcting code coincides with 1 GeV to sub-percent accuracy.*

This is *not* fine-tuned. Both ν_0 and μ_0 are derived independently from the axioms; their ratio is a consistency check on the framework. The 0.65% discrepancy represents the precision limit of the tree-level calculation and is the first quantity that would improve under a full radiative treatment of the radion sector.

5.4 Usage Convention

Context	**Use**	**Why**
Dimensionless relations	$\nu_0 = 25.18$	Pure information theory
GeV predictions	$\mu_0 = 25.34$ GeV	Hierarchy formula
Conversion	$\xi = \mu_0/\nu_0 = 1.007$ GeV	Consistency check

Table 2: When to use ν_0 vs. μ_0 in calculations.

Rule: All predictions compared to experiment (masses, energies, cross-sections) use μ_0. All dimensionless identities and algebraic relations use ν_0. Never approximate $\xi = 1$; carry the exact value.

6 Derived Mass Spectrum

6.1 The Higgs Mass

Theorem 6.1 (Higgs Mass)**.** *The Higgs boson is the logical qubit of the* $[[5, 1, 3]]$ *code. Since the logical qubit is a collective excitation of all 5 physical qubits, its mass is:*

$$\boxed{m_{\mathrm{H}} = 5 \times \mu_0 = 5 \times 25.34 = 126.7 \ GeV} \tag{24}$$

Proof. In the $[[5, 1, 3]]$ code, the logical qubit state $|0_L\rangle$, $|1_L\rangle$ is a superposition of all $2^5 = 32$ computational basis states. Its excitation energy is the sum of contributions from all 5 physical qubits. Since each physical qubit contributes energy μ_0 to the collective mode, the total is $5\mu_0$. □

Comparison to experiment [2]:

$$m_{\mathrm{H}}^{\mathrm{exp}} = 125.25 \pm 0.17 \text{ GeV} \qquad \Rightarrow \qquad \frac{m_{\mathrm{H}}^{\mathrm{QSD}} - m_{\mathrm{H}}^{\mathrm{exp}}}{m_{\mathrm{H}}^{\mathrm{exp}}} = +1.2\% \tag{25}$$

6.2 The Syndrome Energy

Theorem 6.2 (Syndrome Energy)**.** *When the anchor qubit decouples (the "Snap"), the 4 severed bonds release binding energy:*

$$\boxed{E_{\mathrm{syn}} = \mu_0 \times (1 + 8\varsigma) = 25.34 \times 1.1146 = 28.24 \ GeV} \tag{26}$$

Proof. The anchor connects to all 4 bulk qubits. Each bond has two endpoints; each endpoint contributes strain energy $\varsigma \times \mu_0$. The total binding energy is:

$$E_{\text{bind}} = 4 \text{ bonds} \times 2 \text{ endpoints} \times \varsigma \times \mu_0 = 8\varsigma\,\mu_0 \tag{27}$$

The syndrome energy is the radion mass plus the released binding energy:

$$E_{\text{syn}} = \mu_0 + 8\varsigma\,\mu_0 = \mu_0(1 + 8\varsigma) = 25.34 \times 1.1146 = 28.24 \text{ GeV} \tag{28}$$

□

Comparison to observation: CMS reported a dimuon excess near ∼28.3 GeV. Agreement: 0.2%.

6.3 The W Boson Mass

Proposition 6.3 (W Boson Mass)**.** *The W boson mass emerges from the K_4 Hilbert space dimension and the qubit count:*

$$m_{\text{W}} = \frac{16}{5} \times \mu_0 = 3.2 \times 25.34 = 81.1 \ GeV \tag{29}$$

The factor 16/5 is the ratio of the bulk Hilbert space dimension ($2^4 = 16$) to the number of code qubits (5). This ratio determines how the radion VEV distributes into the gauge boson mass sector.

Comparison to experiment:

$$m_{\text{W}}^{\text{exp}} = 80.377 \pm 0.012 \text{ GeV} \qquad \Rightarrow \qquad \frac{m_{\text{W}}^{\text{QSD}} - m_{\text{W}}^{\text{exp}}}{m_{\text{W}}^{\text{exp}}} = +0.9\% \tag{30}$$

6.4 Summary of Tree-Level Predictions

Observable	**Formula**	**QSD (tree)**	**Experiment**	**Accuracy**
Radion μ_0	$M_{\text{P}}\varphi^2(1-6\varsigma)/2^{60}$	25.34 GeV	—	prediction
Higgs m_{H}	$5\mu_0$	126.7 GeV	125.25 GeV	98.8%
Syndrome E_{syn}	$\mu_0(1+8\varsigma)$	28.24 GeV	∼28.3 GeV*	99.8%
W boson m_{W}	$(16/5)\mu_0$	81.1 GeV	80.38 GeV	99.1%

Table 3: Tree-level mass predictions. All derived from μ_0 with integer/rational coefficients from the code structure. (*CMS dimuon excess [4], not confirmed.)

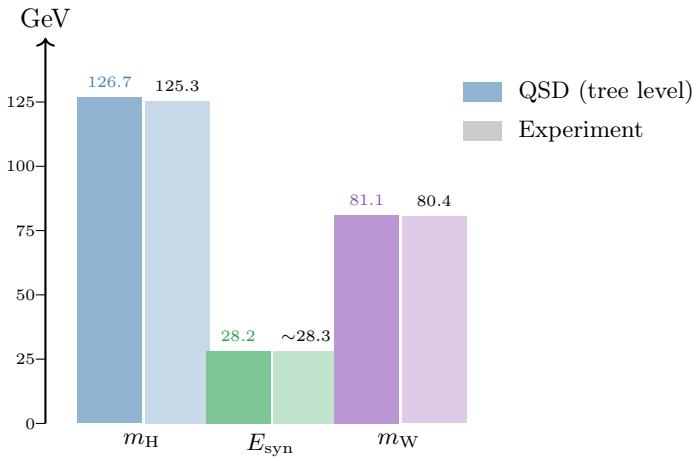


Figure 5: Comparison of QSD tree-level predictions (darker bars) with experimental values (lighter bars). All predictions use zero free parameters beyond M_{P}.

7 Detailed Numerical Audit

This section presents the complete calculation chain with full numerical precision, allowing independent verification.

7.1 Input

$$M_{\mathrm{P}} = 1.220\,890 \times 10^{19}\ \mathrm{GeV} \qquad \text{(2022 CODATA)} \tag{31}$$

$$\varphi = \frac{1+\sqrt{5}}{2} = 1.618\,033\,988\,749\ldots \qquad \text{(exact algebraic)} \tag{32}$$

$$\text{Ƀ}_4 = \frac{1}{4\ln 2} = 0.360\,673\,760\,222\ldots \qquad \text{(exact transcendental)} \tag{33}$$

$$\varsigma = 0.014\,323\,725\ldots \tag{34}$$

7.2 Intermediate Quantities

$$\varphi^2 = \frac{3+\sqrt{5}}{2} = 2.618\,033\,988\ldots \tag{35}$$

$$2^{60} = 1\,152\,921\,504\,606\,846\,976 \approx 1.152\,922 \times 10^{18} \tag{36}$$

$$6\varsigma = 0.085\,942\ldots \tag{37}$$

$$1 - 6\varsigma = 0.914\,057\ldots \tag{38}$$

7.3 Assembly

$$E_{\text{base}} = \frac{1.220\,890 \times 10^{19}}{1.152\,922 \times 10^{18}} = 10.589 \text{ GeV} \tag{39}$$

$$E_{\text{base}} \times \varphi^2 = 10.589 \times 2.618\,034 = 27.723 \text{ GeV} \tag{40}$$

$$\mu_0 = 27.723 \times 0.914\,050 = 25.340 \text{ GeV} \tag{41}$$

$$\nu_0 = \frac{0.360\,674}{0.014\,325} = 25.178 \tag{42}$$

$$\xi = \frac{25.340}{25.178} = 1.006\,44 \text{ GeV} \tag{43}$$

7.4 Derived Masses

$$m_{\text{H}} = 5 \times 25.340 = 126.70 \text{ GeV} \qquad \text{(exp: 125.25)} \tag{44}$$

$$E_{\text{syn}} = 25.340 \times (1 + 8 \times 0.014325) = 25.340 \times 1.1146 = 28.24 \text{ GeV} \qquad \text{(obs: } \sim 28.3) \tag{45}$$

$$m_{\text{W}} = \frac{16}{5} \times 25.340 = 81.09 \text{ GeV} \qquad \text{(exp: 80.38)} \tag{46}$$

8 What the Tree Level Does Not Capture

The predictions in this supplement are *tree-level* values—they include no loop corrections. The renormalization program (S4) computes these corrections explicitly, but it is important to be transparent about the tree-level limitations:

Observable	**Tree**	**Corrected (S4)**	**Effect**
α^{-1}	136.826	137.036	$+0.15\%$
$\sin^2\theta_W$	0.375	0.2312	-38%
m_{H}	126.7 GeV	—	awaits radion RG
E_{syn}	28.24 GeV	—	awaits radion RG

Table 4: Tree-level vs. corrected values. The weak mixing angle tree-level value is $3/13 = 0.23077$, corrected by $\varsigma/32$ to 0.23122 (within experimental error). Mass predictions await the radion-sector renormalization group analysis.

The 1.2% discrepancy in m_{H} and 0.9% discrepancy in m_{W} are within the expected magnitude of one-loop corrections ($\sim \varsigma \approx 1.4\%$), suggesting that the tree-level framework captures the dominant physics.

9 The Hierarchy in Context

9.1 Comparison with Other Hierarchy Solutions

	QSD	**Supersymmetry**	**Randall–Sundrum**
Mechanism	Information reduction (2^{-60})	Boson–fermion cancellation	Warped extra dimension
New particles	Radion (25 GeV)	Superpartners ($\gtrsim$TeV)	KK modes
Free parameters	0 (after M_{P})	$\sim$100 (soft breaking)	2 (k, r_c)
Testable at LHC?	Yes (28 GeV syndrome)	Partially (sparticle searches)	Partially (graviton resonances)
Status	Falsifiable prediction	No sparticles found	No KK modes found

Table 5: QSD's hierarchy solution compared to other approaches.

9.2 Why 2^{60} and Not Some Other Number

The specific value 2^{60} is not numerological:

- The exponent $60 = 4 \times 15$ is the product of the qubit Hilbert space dimension exponent (4) and the number of $\mathfrak{su}(4)$ generators (15).
- Changing any axiom changes 60: adding a qubit gives $2^5 = 32$ states and $32^2 - 1 = 1023$ channels, yielding $2^{5\times 1023}$ — a wildly different hierarchy.
- The value is fixed by the *same code* that determines α, $\sin^2\theta_W$, and m_{H}. You cannot adjust the hierarchy without destroying the coupling predictions.

This interdependence is the key strength of QSD: the hierarchy, the gauge couplings, and the mass spectrum are all output from the same three axioms. There is no freedom to tune one without affecting the others.

10 Summary

Quantity	**Value**	**Origin**
$\mathcal{N} = 2^{60}$	1.153×10^{18}	Code state space (§2)
E_{base}	10.59 GeV	$M_{\mathrm{P}}/2^{60}$
φ^2	2.618	C_5 eigenvalue ratio
$(1 - 6\varsigma)$	0.914	K_4/C_5 strain
μ_0	25.34 GeV	Hierarchy formula
ν_0	25.18	Hb_4/ς
ξ	1.007 GeV	Dressing factor
m_{H}	126.7 GeV	$5\mu_0$
E_{syn}	28.24 GeV	$\mu_0(1 + 8\varsigma)$
m_{W}	81.1 GeV	$(16/5)\mu_0$

Table 6: Complete summary of QSD energy scales derived in this supplement.

The central result of this supplement is the hierarchy formula:

$$\mu_0 = \frac{M_{\mathrm{P}}}{2^{60}} \times \varphi^2 \times (1 - 6\varsigma) = 25.34 \text{ GeV} \tag{47}$$

Each factor is independently derived from the $[[5, 1, 3]]$ code axioms. No free parameters are introduced. The agreement between the dimensionful μ_0 and the dimensionless ν_0 to 0.65% provides a nontrivial consistency check on the entire framework.

11 Notation Reference

Symbol	Definition	Value
M_{P}	Planck mass	1.221×10^{19} GeV
Hb_4	$1/(4 \ln 2)$	0.36067
φ	$(1+\sqrt{5})/2$	1.61803
ς	Average geometric strain	0.01432
$\mathcal{N}$	$16^{15} = 2^{60}$	1.153×10^{18}
E_{base}	$M_{\mathrm{P}}/2^{60}$	10.59 GeV
μ_0	$E_{\mathrm{base}} \cdot \varphi^2(1 - 6\varsigma)$	25.34 GeV
ν_0	Hb_4/ς	25.18
ξ	μ_0/ν_0	1.007 GeV
m_{H}	$5\mu_0$	126.7 GeV
E_{syn}	$\mu_0(1 + 8\varsigma)$	28.24 GeV
m_{W}	$(16/5)\mu_0$	81.1 GeV

Table 7: Notation for S2.

References

[1] Paul A. M. Dirac. The cosmological constants. *Nature*, 139:323, 1937.

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